Bohmian quantum field theory

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- Non-relativistic Bohmian mechanics
  - Point-particles with positions $\mathbf{X}_1, \ldots, \mathbf{X}_n$
  - Dynamics depends on the wave function ψ

$$\frac{d\mathbf{X}_i}{dt} = \mathbf{v}_i^{\psi}(\mathbf{X}_1, \ldots, \mathbf{X}_n)$$

  - ψ satisfies non-relativistic Schrödinger equation

$$\mathrm{i}\partial_t \psi = H_{\mathrm{n.r.}}\psi$$

- Bohmian quantum field theory
  - Point-particles or fields?
  - Dynamics that depends on ψ
  - ψ satisfies Schrödinger equation

$$\mathrm{i}\partial_t \psi = H_{\mathrm{QFT}}\psi$$

Non-relativistic Bohmian mechanics

(a.k.a. pilot-wave theory, de Broglie-Bohm theory, . . .)

- De Broglie (1927), Bohm (1952)




- Point-particles moving under influence of the wave function.
- Positions of the particles: $\mathbf{X}_1, \ldots, \mathbf{X}_n$

  Configuration $X = (\mathbf{X}_1, \ldots, \mathbf{X}_n)$

- Dynamics:

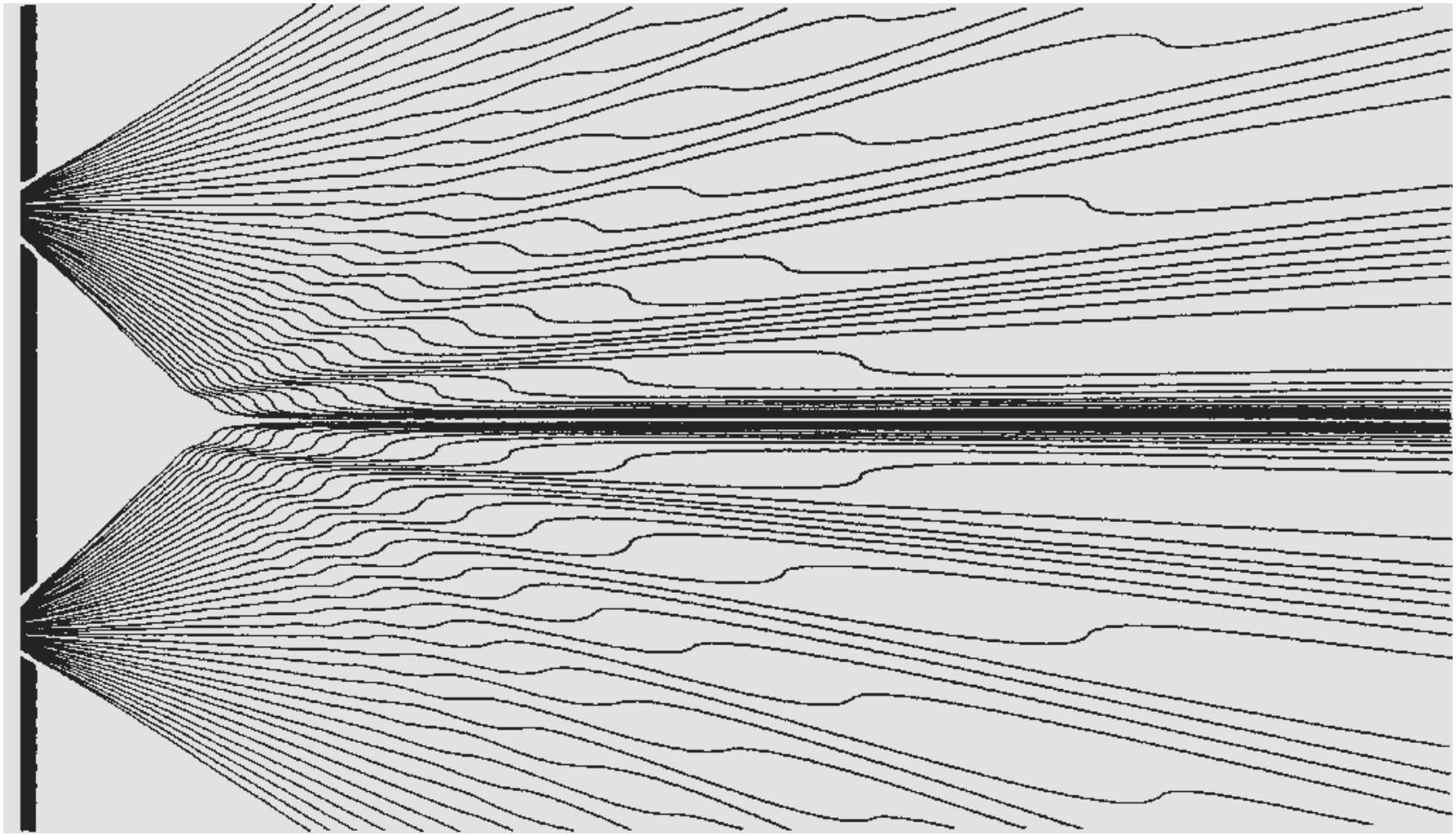
$$\frac{dX(t)}{dt} = v^{\psi_t}(X(t))$$

where

$$v^\psi = (\mathbf{v}_1^\psi, \ldots, \mathbf{v}_n^\psi), \qquad \mathbf{v}_k^\psi = \frac{\hbar}{m_k}\mathrm{Im}\frac{\boldsymbol{\nabla}_k\psi}{\psi} = \frac{1}{m_k}\boldsymbol{\nabla}_k S, \qquad \psi = |\psi|e^{\mathrm{i}S/\hbar}$$

$$\mathrm{i}\hbar\partial_t\psi_t(x) = \left(-\sum_{k=1}^{N}\frac{\hbar^2}{2m_k}\nabla_k^2 + V(x)\right)\psi_t(x)\,, \qquad x = (\mathbf{x}_1, \ldots, \mathbf{x}_N)$$

- Double Slit experiment:



- Quantum equilibrium and standard quantum predictions
  - Quantum equilibrium:
    - for an ensemble of systems with wave function ψ
    - equilibrium distribution of particle positions $\rho(x) = |\psi(x)|^2$
  - Quantum equilibrium is preserved by the particle motion (= equivariance), i.e.

$$\rho(x, t_0) = |\psi(x, t_0)|^2 \qquad \Rightarrow \qquad \rho(x, t) = |\psi(x, t)|^2 \qquad \forall t$$

    because guidance equation implies

$$\partial_t \rho + \sum_{k=1}^{n} \boldsymbol{\nabla}_k \cdot (\mathbf{v}_k^{\psi} \rho) = 0$$

    and Schrödinger equation implies

$$\partial_t |\psi|^2 + \sum_{k=1}^{n} \boldsymbol{\nabla}_k \cdot (\mathbf{v}_k^{\psi} |\psi|^2) = 0$$

  - Agreement with quantum theory in quantum equilibrium. We have quantum equilibrium in a typical universe (Dürr, Goldstein, Zanghì).

- Recipe to construct Bohmian theories for a given quantum theory
  - Choose ontology in space(-time), also called local beables, say particles X
  - Consider the quantum distribution $|\psi|^2$
  - Consider the continuity equation for $|\psi|^2$:

$$\partial_t |\psi|^2 + \mathrm{div} j^\psi = 0$$

  - Define guidance equation

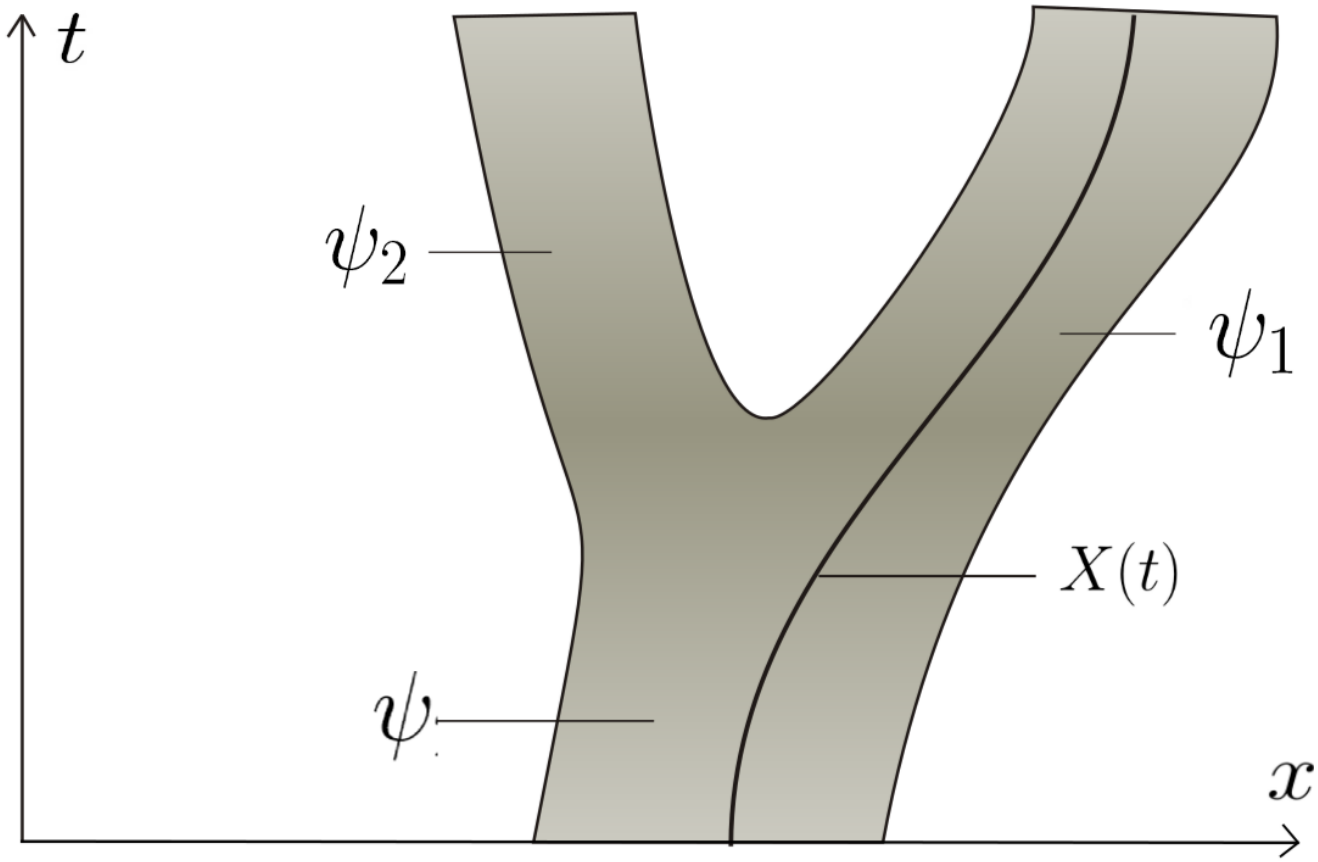
$$\dot{X} = \frac{j^\psi}{|\psi|^2}$$

This receipe will guarantee equivarance of the equilibrium distribution.

But does not necessarily guarantee an empirically adequate theory:

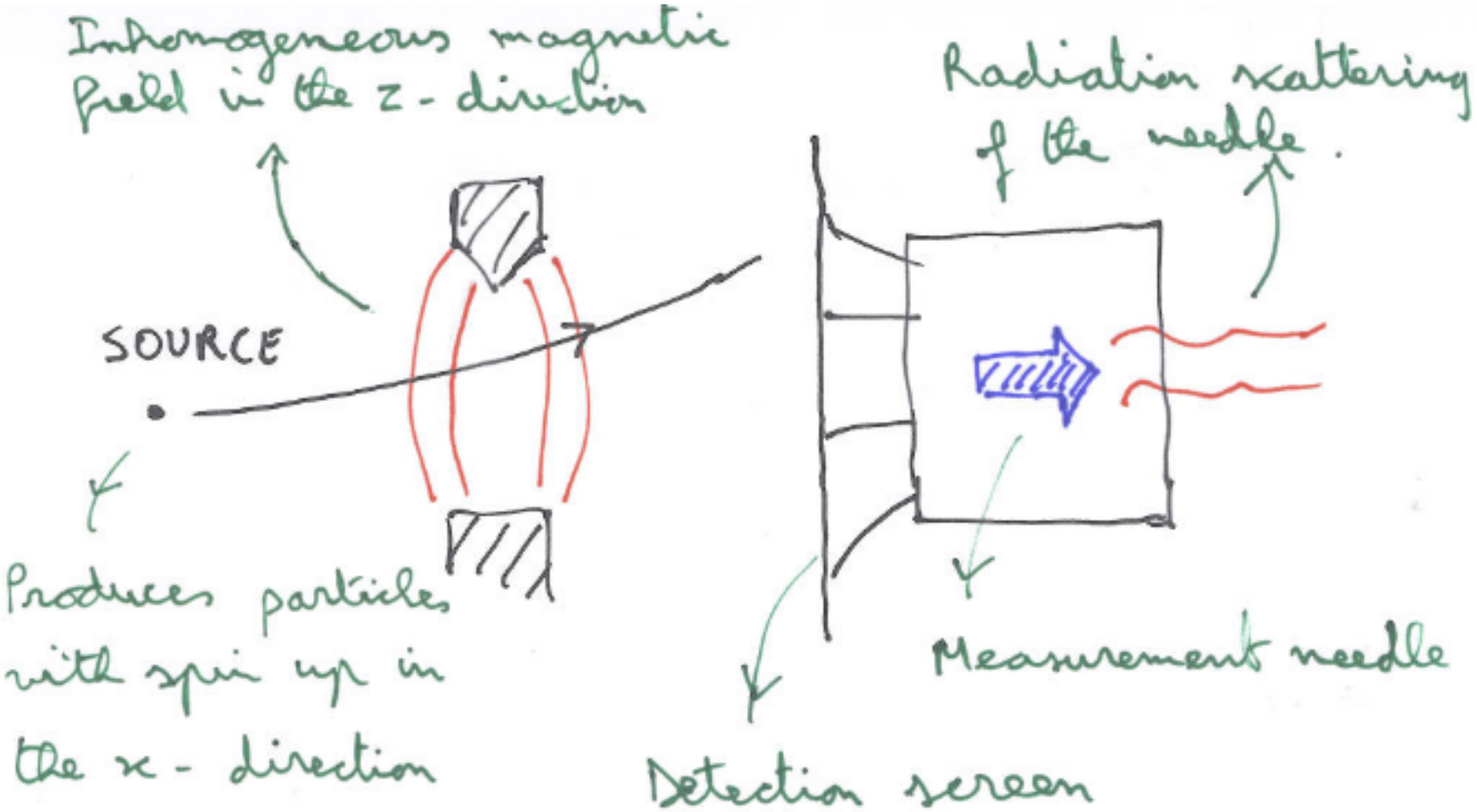
The variables X should contain, on the macroscopic level, an image of the everyday classical world (Bell)

- **Effective collapse of the wave function**
  - Branching of the wave function: $\psi \to \psi_1 + \psi_2 \qquad \psi_1\psi_2 = 0$
  - Effective collapse $\psi \to \psi_1$ (ψ_2 does no longer effect the motion of the configuration X)

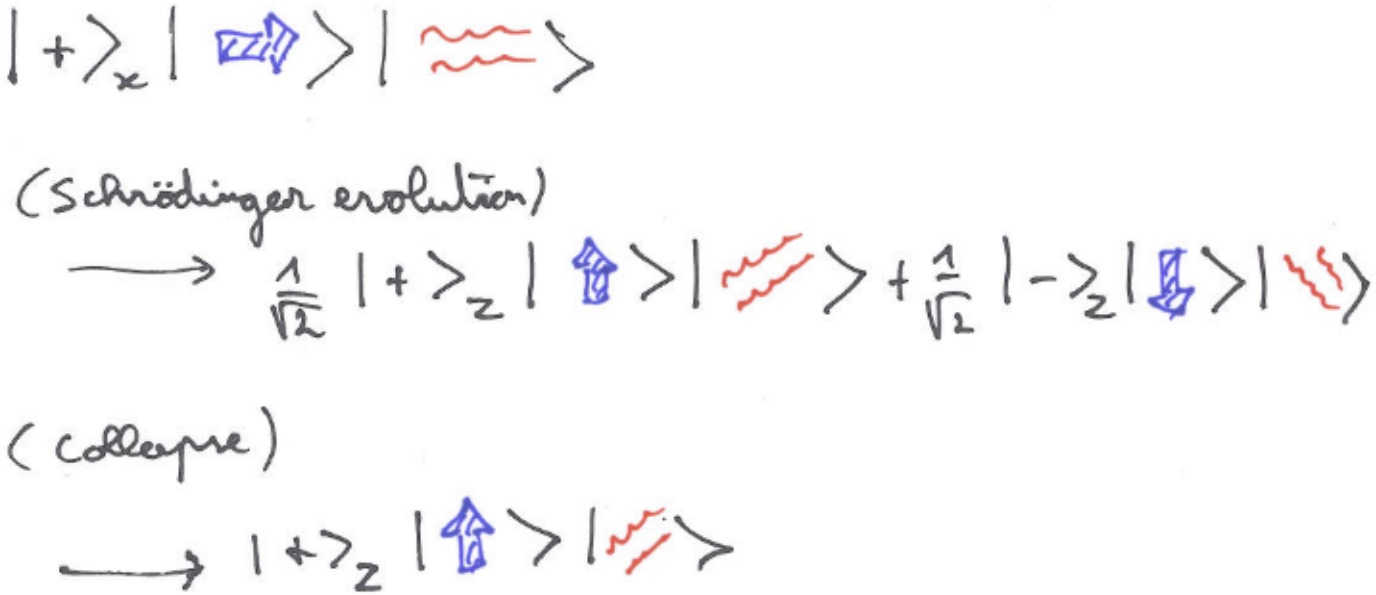


- **Stern-Gerlach experiment**

  Sketch of the setup:



Orthodox quantum description:



Bohmian treatment of spin

– No need to introduce extra variables representing spin.

Just positions:

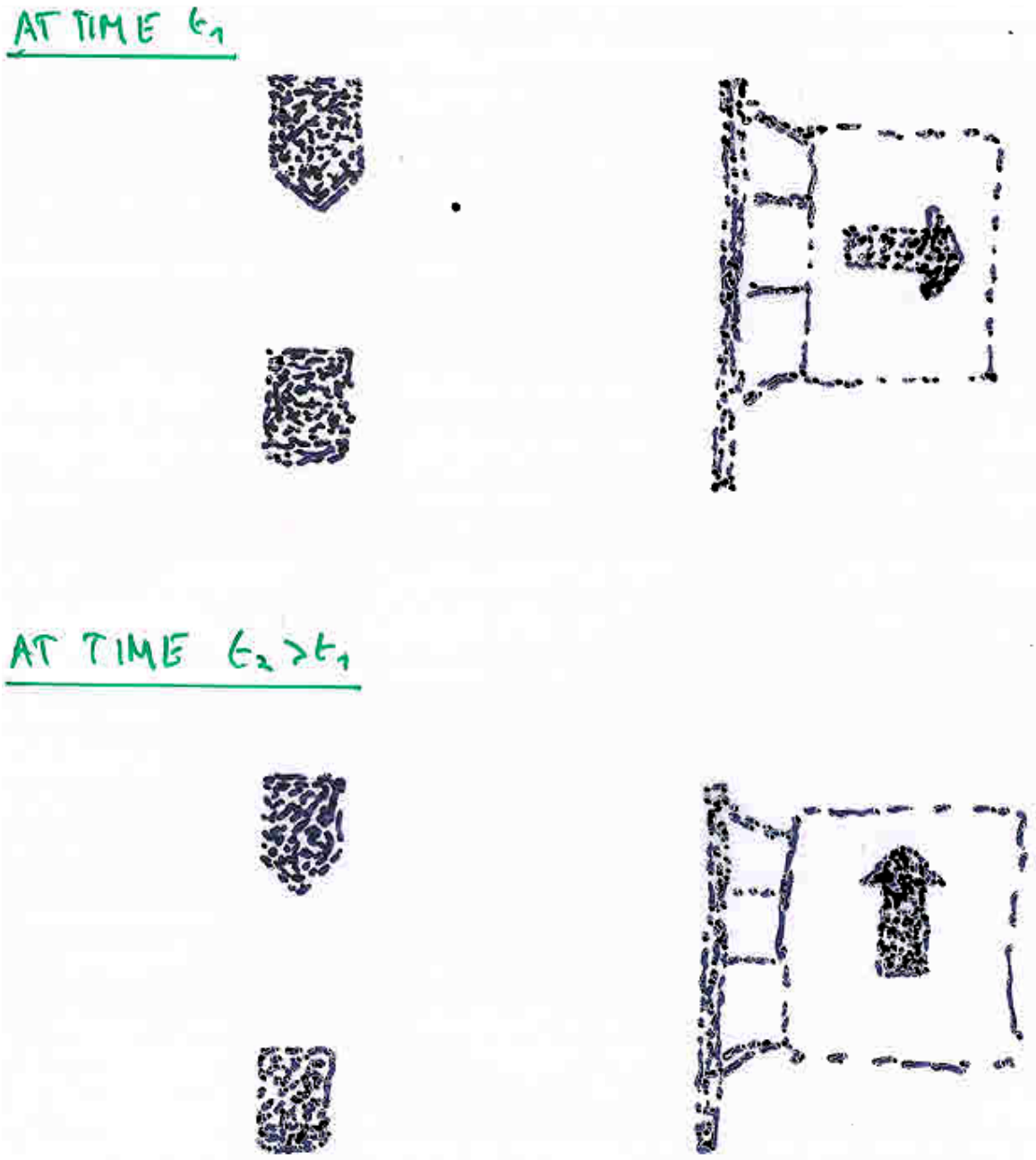


Guidance law for non-relativistic spin-1/2 particles:

$$\frac{d\mathbf{X}_k}{dt} = \frac{\hbar}{m_k}\mathrm{Im}\frac{\psi^\dagger \boldsymbol{\nabla}_k \psi}{\psi^\dagger \psi}, \tag{1}$$

where ψ is Pauli spinor.

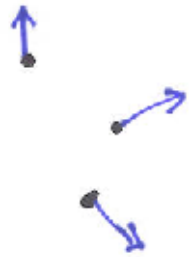
Bohmian picture:



One could introduce extra variables. E.g. the spin vector:

$$\mathbf{s}(t) = \frac{\hbar}{2} \frac{\psi^\dagger(\mathbf{x},t)\boldsymbol{\sigma}\psi(\mathbf{x},t)}{\psi^\dagger(\mathbf{x},t)\psi(\mathbf{x},t)}\Big|_{\mathbf{x}=\mathbf{X}(t)}$$

which is constructed from the Pauli spinor ψ and the actual position of the particle.



- **Dirac theory (first-quantized level)**
  - Single particle

    Dirac equation:

    $$\mathrm{i}\partial_\mu \gamma^\mu \psi - m\psi = 0$$

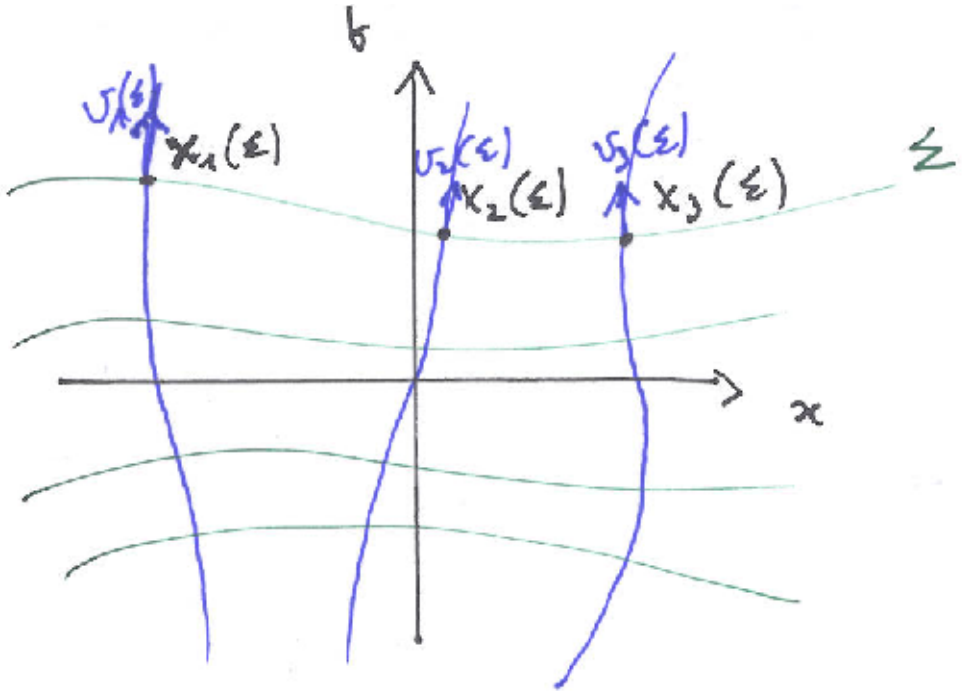
    Guidance equation:

    $$\frac{dX^\mu}{d\tau} \sim j^\mu = \psi^\dagger \gamma^0 \gamma^\mu \psi$$

    $\rightarrow$ Is fully Lorentz invariant

– Many particles

Dynamics depends on particular reference frame or foliation $\mathcal{F}$ (to guarantee equivariance): $\dot{X}_k(\Sigma) = v_k^{\psi,\mathcal{F}}(X_1(\Sigma), \ldots, X_N(\Sigma))$



This dependence makes the theory not Lorentz invariant. Nevertheless the preferred frame or foliation can not be detected. The statistical predictions are Lorentz invariant.

Dynamics can be made Lorentz invariant by a covariant determination of the foliation by the wave function: $\mathcal{F} = \mathcal{F}^{\psi}$ (Dürr, Goldstein, Norsen, Struyve, Zanghì 2014)

Bohmian quantum field theory

- Need suitable variables (local beables). Particles, fields?
- Need wave function and Schrödinger evolution.
- We assume suitable regulators which make equations well-defined.
- We assume preferred reference frame.

Field ontology

Bosonic fields

- Scalar field (Bohm (1952))

  Hamiltonian:

$$\widehat{H} = \frac{1}{2}\int d^3x \left(\widehat{\Pi}^2 + (\nabla\widehat{\phi})^2 + m^2\widehat{\phi}^2\right), \qquad [\widehat{\phi}(\mathbf{x}), \widehat{\Pi}(\mathbf{y})] = \mathrm{i}\delta(\mathbf{x}-\mathbf{y})$$

  Functional Schrödinger representation:

$$\widehat{\phi}(\mathbf{x}) \to \phi(\mathbf{x}), \qquad \widehat{\pi}(\mathbf{x}) \to -\mathrm{i}\frac{\delta}{\delta\phi(\mathbf{x})}$$

$$\mathrm{i}\frac{\partial\Psi(\phi,t)}{\partial t} = \frac{1}{2}\int d^3x \left(-\frac{\delta^2}{\delta\phi^2} + (\nabla\phi)^2 + m^2\phi^2\right)\Psi(\phi,t).$$

  Bohmian field $\phi(\mathbf{x})$ with guidance equation:

$$\frac{\partial\phi(\mathbf{x},t)}{\partial t} = \frac{\delta S(\phi,t)}{\delta\phi(\mathbf{x})}\Big|_{\phi=\phi(\mathbf{x},t)}, \qquad \Psi = |\Psi|e^{\mathrm{i}S}$$

- Similarly for other bosonic fields (see Struyve (2010) for a review).

  For example, electromagnetic field: $(A(\mathbf{x}), \Psi(A))$

Fermionic fields

- Valentini (1992)
  - Wave function $\Psi(\eta)$,

    $\eta(\mathbf{x})$ is a Grassmannian field,

    Ψ is an element of a Grassmann algebra (not $\mathbb{C}$)
  - Local beable: $\eta(\mathbf{x})$
  - Problem: What is the equilibrium distribution? What is an adequate guidance equation?

- Holland (1988,1993)
  - Wave function $\Psi(\boldsymbol{\alpha})$,

    $\boldsymbol{\alpha}(\mathbf{k}) = (\alpha(\mathbf{k}), \beta(\mathbf{k}), \gamma(\mathbf{k}))$ set of Euler angles for each point in momentum space

Fermionic fields

- Holland (1988,1993)
  - Wave function $\Psi(\boldsymbol{\alpha})$,

    $\boldsymbol{\alpha}(\mathbf{k}) = (\alpha(\mathbf{k}), \beta(\mathbf{k}), \gamma(\mathbf{k}))$ set of Euler angles for each point in momentum space
  - Local beable: Isn't specified by Holland, but could be constructed from beables $\boldsymbol{\alpha}(\mathbf{k})$ and the wave function.
  - Hard to see whether it is empirically adequate
- Variant of Holland's approach, Struyve (2007)
  - Wave function $\Psi(\boldsymbol{\alpha})$,

    $\boldsymbol{\alpha}(\mathbf{x}) = (\alpha(\mathbf{x}), \beta(\mathbf{x}), \gamma(\mathbf{x}))$ set of Euler angles for each point in physical space
  - Local beable: $\boldsymbol{\alpha}(\mathbf{x})$
  - Seems empirically inadequate!

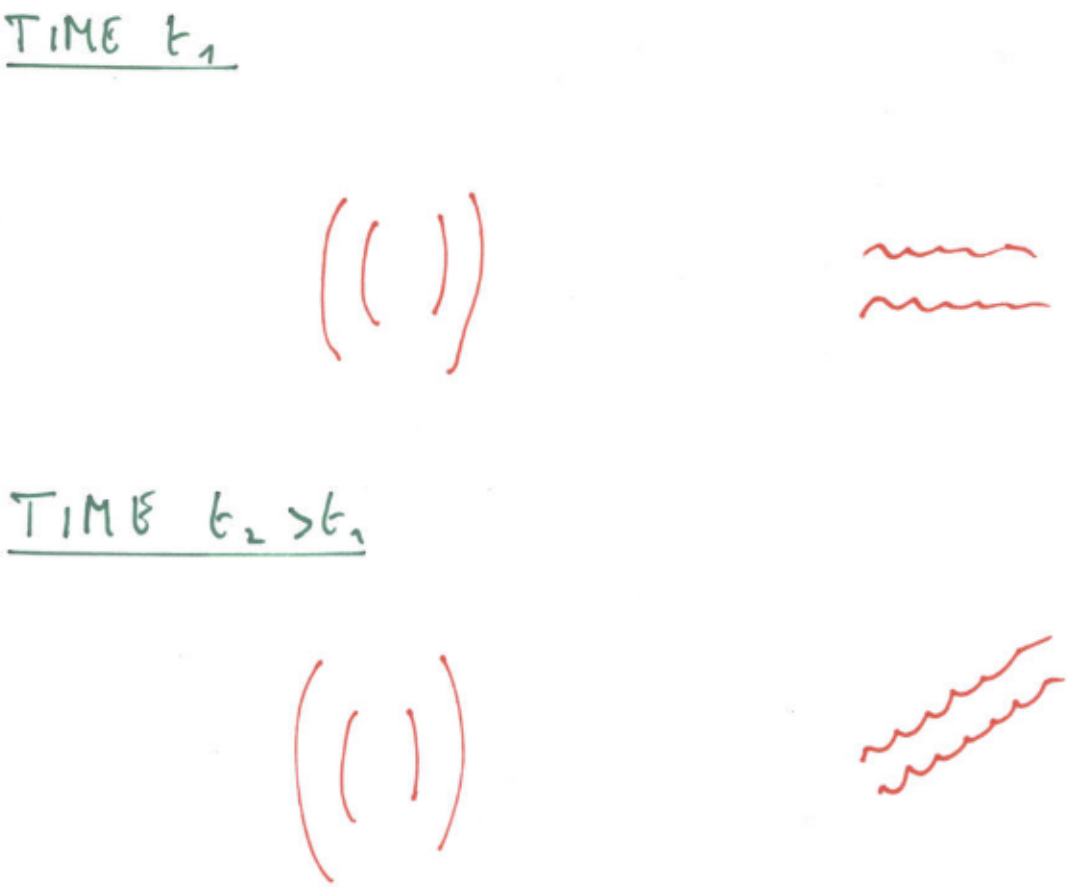
Stern-Gerlach experiment:



Minimalist model

- Struyve & Westman (2006,2007)
  - Wave function: $\Psi_f(A) = \langle f, A|\Psi\rangle$,
    $\rightarrow$ f labels the fermionic degrees of freedom
    $\rightarrow$ $A(\mathbf{x})$ is the vector potential
  - Local beable: $A(\mathbf{x})$
    $\rightarrow$ None for fermionic degrees of freedom
  - Equilibrium distribution: $\sum_f |\Psi_f(A)|^2$

Stern-Gerlach experiment:



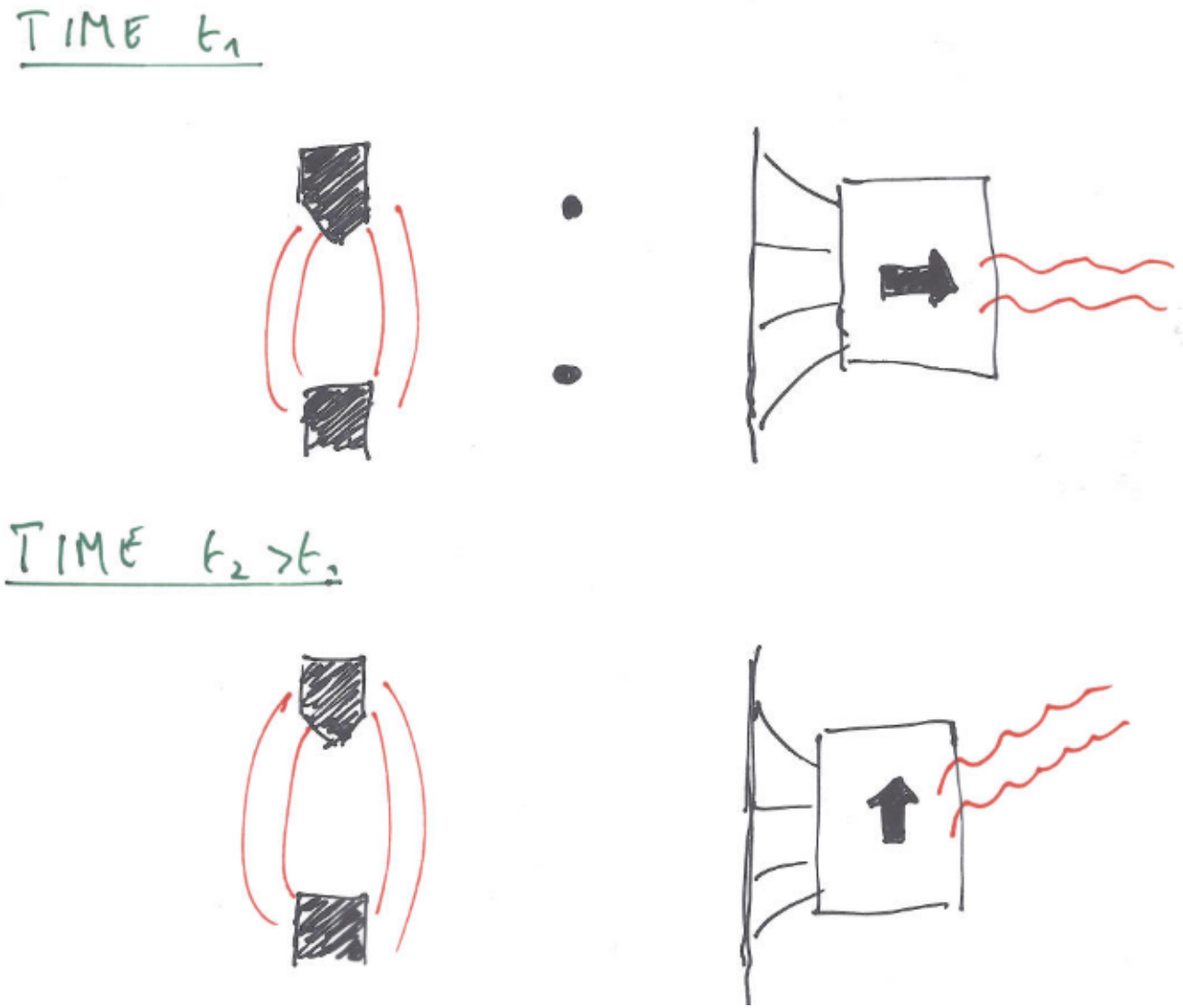
Minimalist model + extras

- Struyve & Westman (2007)
  - Wave function: $\Psi_f(A) = \langle f, A|\Psi\rangle$,
  - Local beables:
    - $\rightarrow$ potential $A(\mathbf{x})$
    - $\rightarrow$ charge distribution $\rho(\mathbf{x})$, given by

$$\rho(\mathbf{x},t) = \left.\frac{\sum_{f,f'} \Psi_f^*(A,t)\widehat{\rho}_{ff'}(\mathbf{x})\Psi_{f'}(A,t)}{\sum_f |\Psi_f(A,t)|^2}\right|_{A=A(t)}$$

where $\widehat{\rho}_{ff'}(\mathbf{x}) = \langle f|\widehat{\rho}(\mathbf{x})|f'\rangle$, with $\widehat{\rho}(\mathbf{x})$ the charge density operator

Stern-Gerlach experiment:



Particle ontology

- Bell (1984)
  - Considers spatial lattice
  - Local beables: fermion numbers $n = (n_1, \ldots, n_l)$ at the lattice points
  - Stochastic dynamics:

    Transition rates to jump from configuration m to configuration n:

$$T_{nm} = \frac{J^+_{nm}}{P_m}$$

    where $J^+ = \max(J, 0)$, with

$$J_{nm} = \sum_{qp} 2\mathrm{Re}\langle\psi(t)|nq\rangle\langle nq| - \mathrm{i}H|mp\rangle\langle mp|\psi(t)\rangle$$

    and

$$P_m = \sum_q |\langle mq|\psi(t)|^2$$

- Continuum generalizations of Bell's model:

  1. Dürr, Goldstein, Tumulka, Zanghì (2002-2005)

     → Local beables: Particle positions

     → Explicit particle creation and annihilation

     → Stochastic model: There is deterministic motion interrupted by particle creation and annihilation which are stochastic

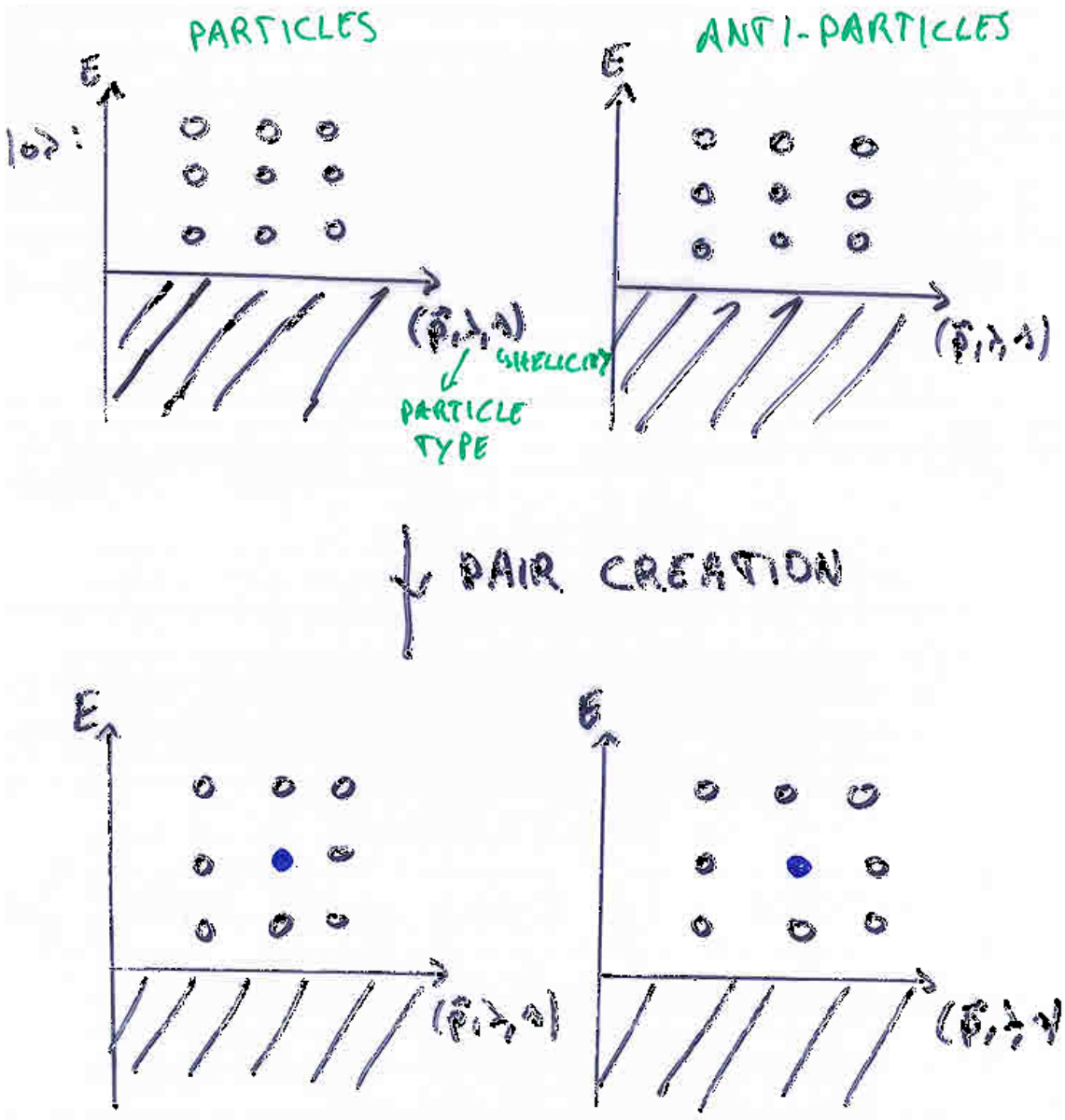
  2. Colin & Struyve (2007)

     → Local beables: Particle positions

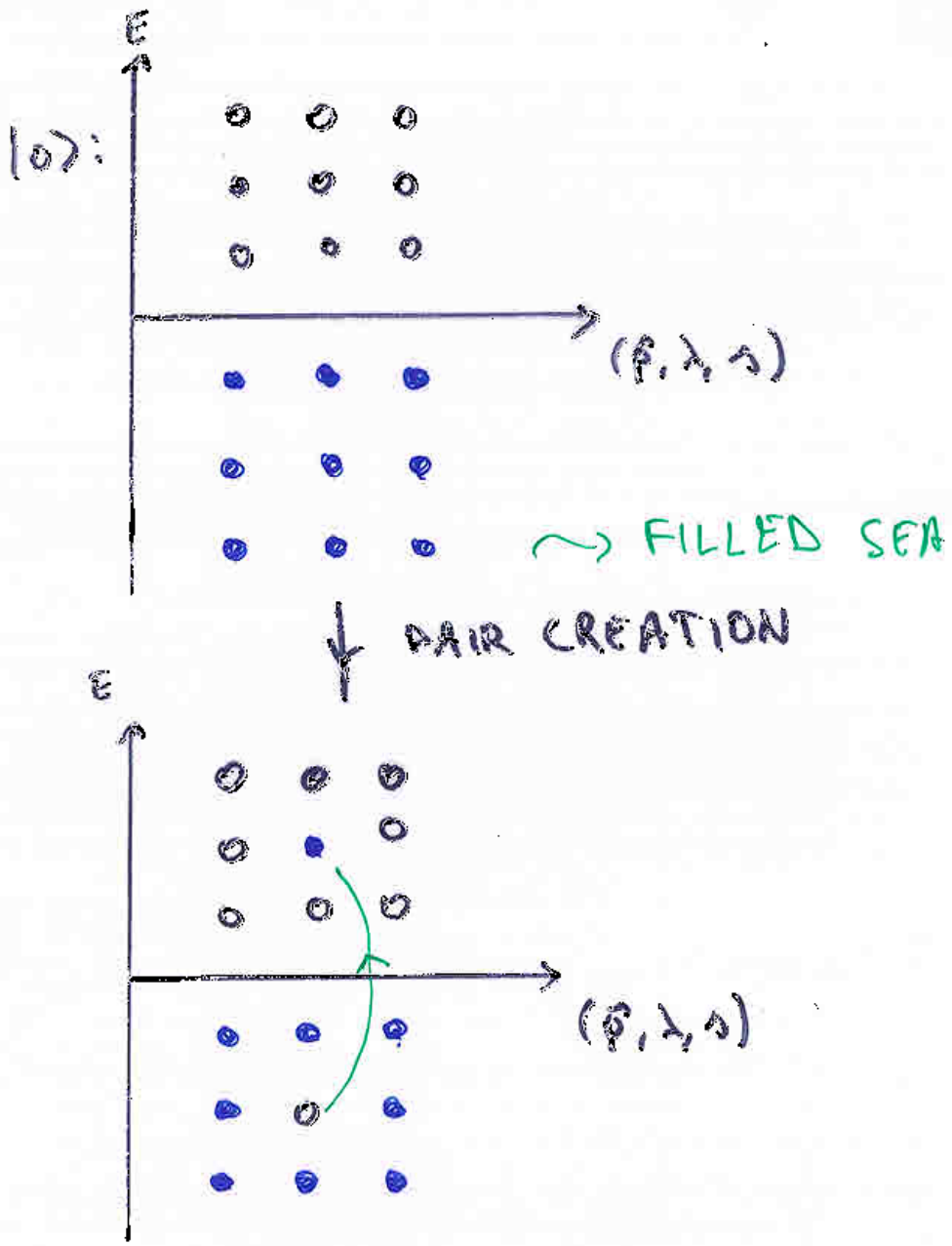
     → Dirac sea picture (number of particles does not change)

     → Deterministic

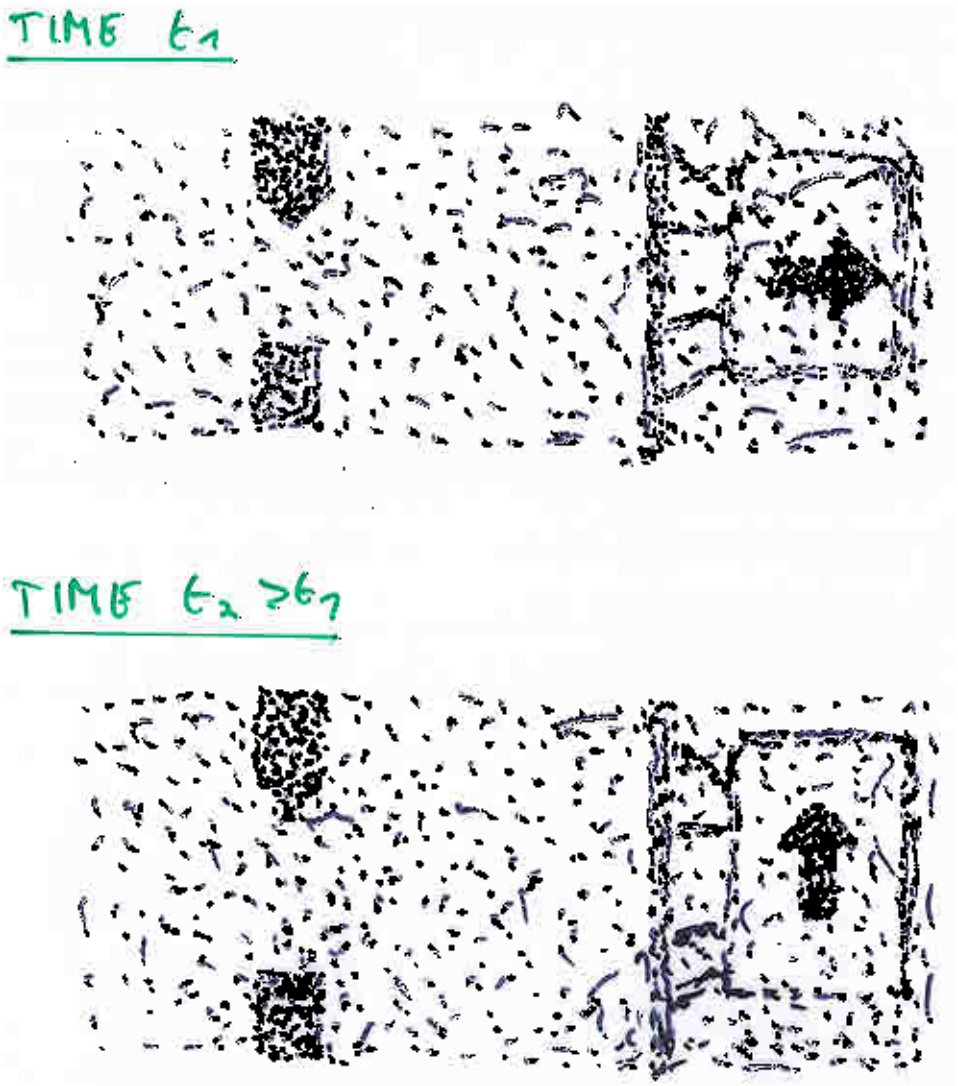
Pair creation in the standard picture:



Pair creation in the Dirac sea picture:



Stern-Gerlach experiment in the Dirac sea - pilot-wave picture:



Conclusion

- In Bohmian QFT:
  - particles positions seem natural for fermions
  - fields seem natural for bosons
- Two candidates for fermions:
  - stochastic one based on the particle–anti-particle picture of QFT
  - deterministic one based on the Dirac sea picture
- The Bohmian theories reproduce the predictions of regularized quantum field theory. The regularization could be understood as a way of dealing with as yet unknown physics at high energies.

  Perhaps the regularization can be removed while preserving a well-defined dynamics for the local beables (even though the wave function evolution might no longer be well-defined).